



NPTEL Course on “Computer System Optimization for ML”

Lecture #5



Agenda

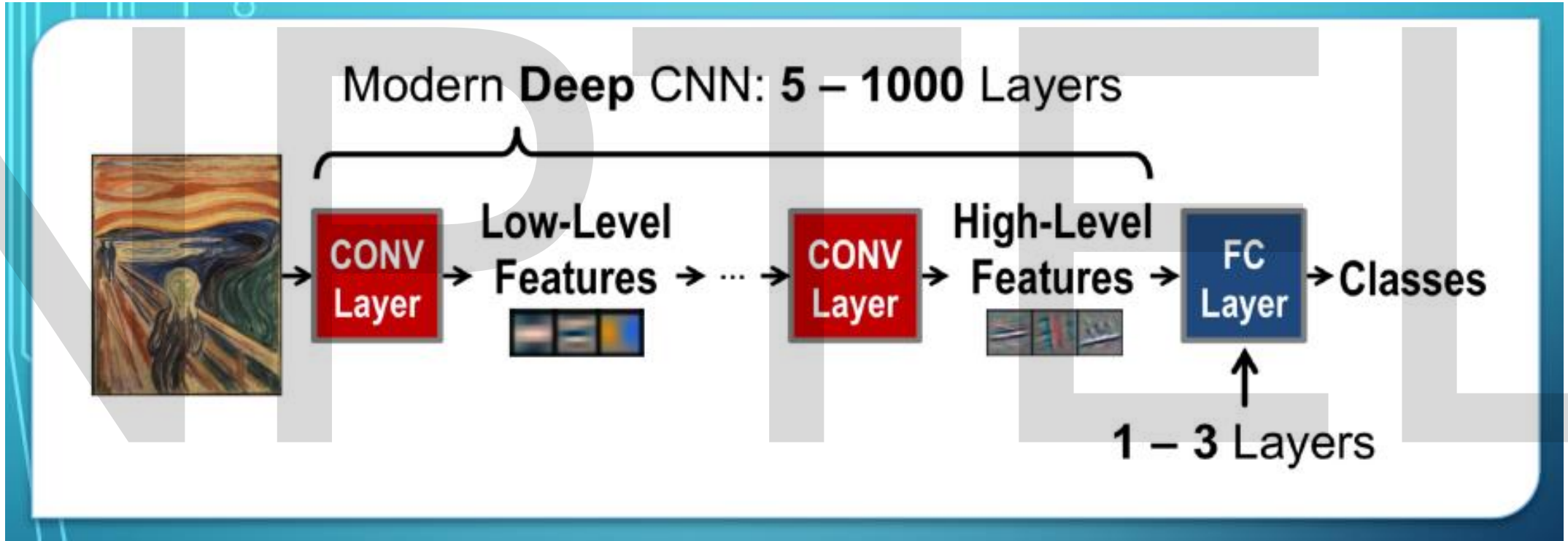
- **Recap**
- **Convolution computation**

NPTREL

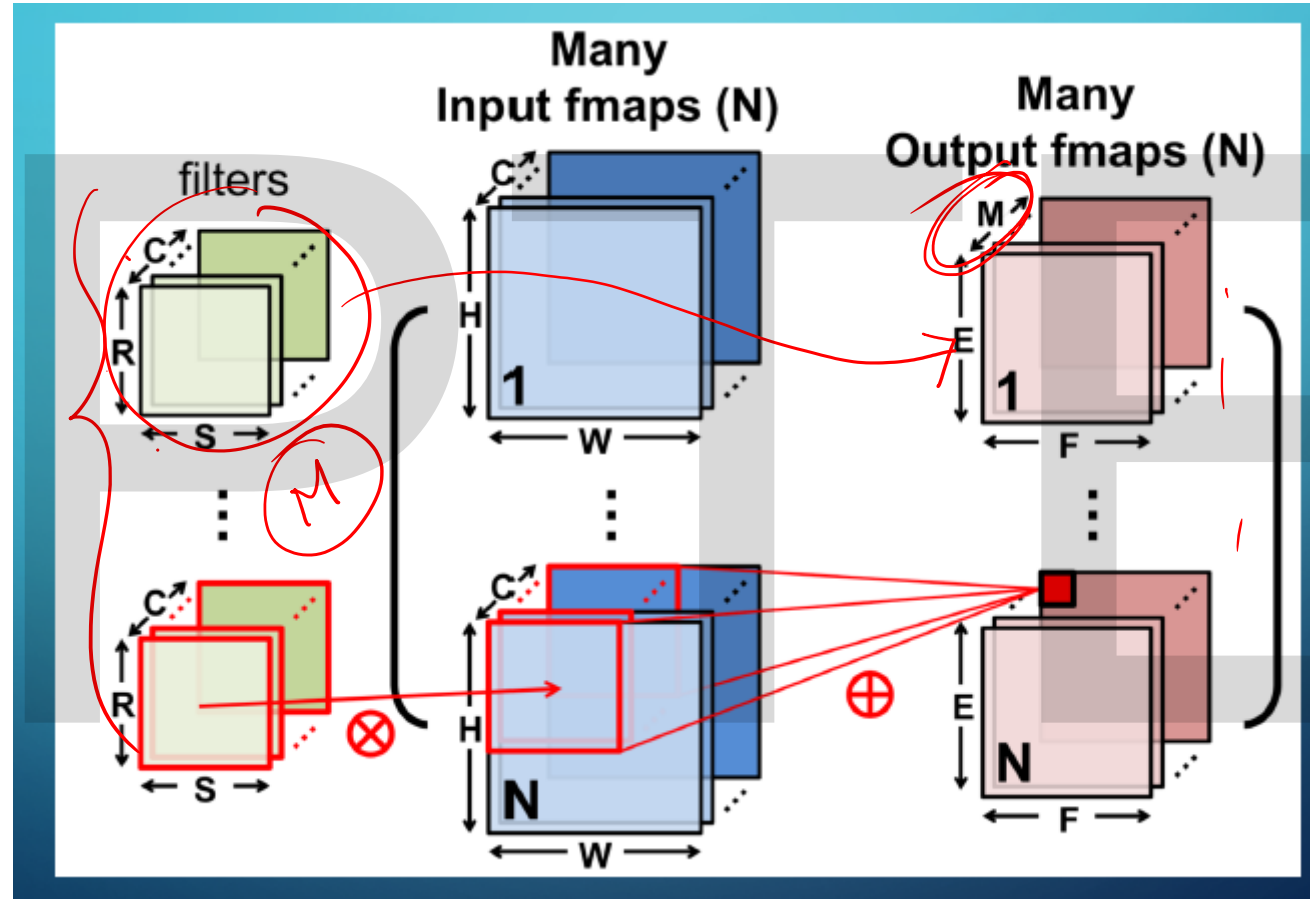
Convolutional Neural Networks (CNNs)

- **DNN treats each pixel in an image as input to a separate neuron**
 - **But it misses the spatial similarity within an image**
- **Why not capture information from the neighbouring pixels**
 - **Convolution!**
- **Convolutions**
 - **Slide a small matrix of parameters (known as filter) across the input image**
 - **Train the filter using backpropagation**

Deep CNNs

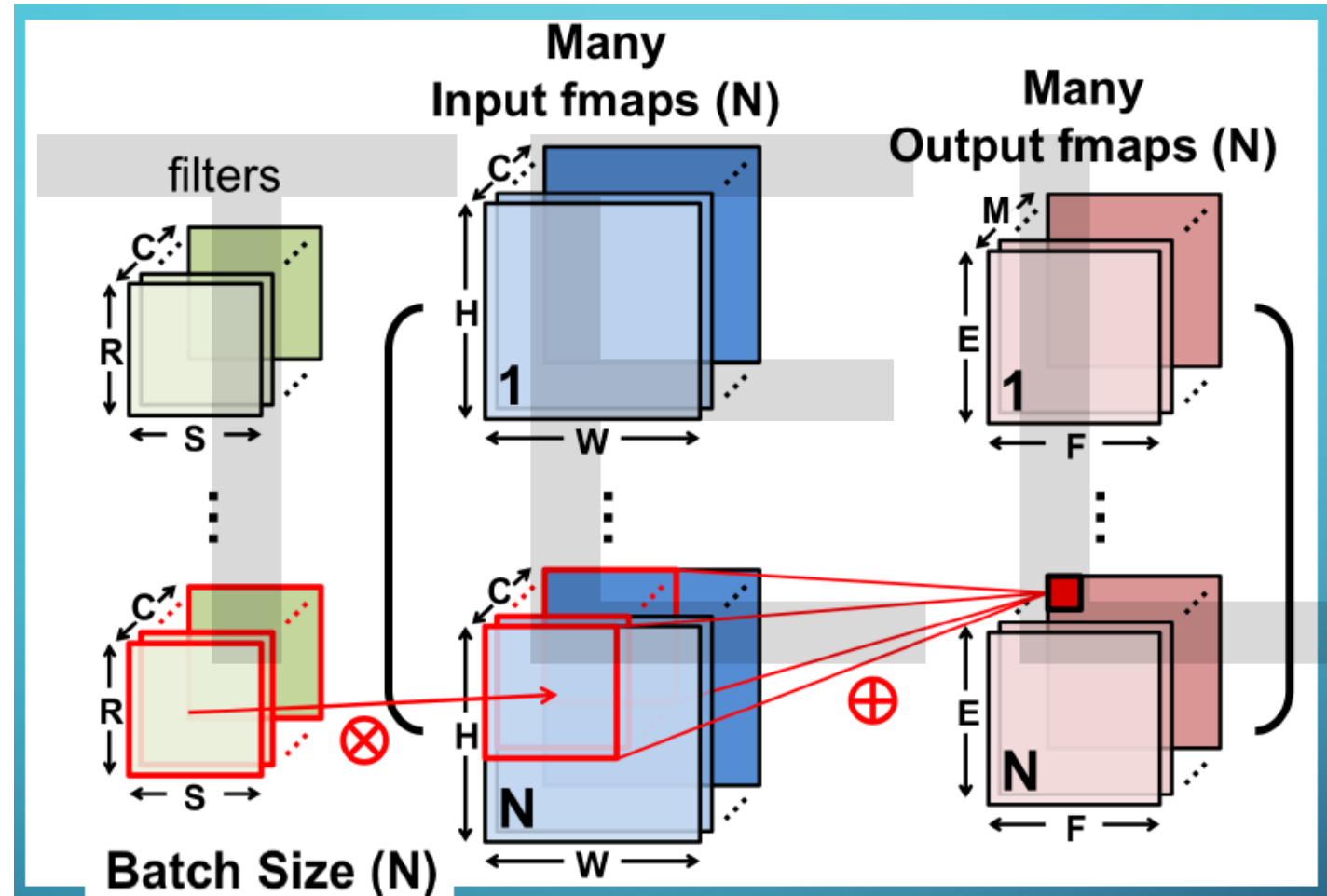


Conv Layer (Batch Multiple Inputs)



Conv Layer

- **Filters are 4-dimensional**
 - $R \times S \times C \times M$
- **What are R, S, C, M?**
 - R: Height of the filter
 - S: Width of the filter
 - C: Number of channels
 - M: Number of filters



Conv Layer Implementation

- Naïve 7-layer for-loop implementation:

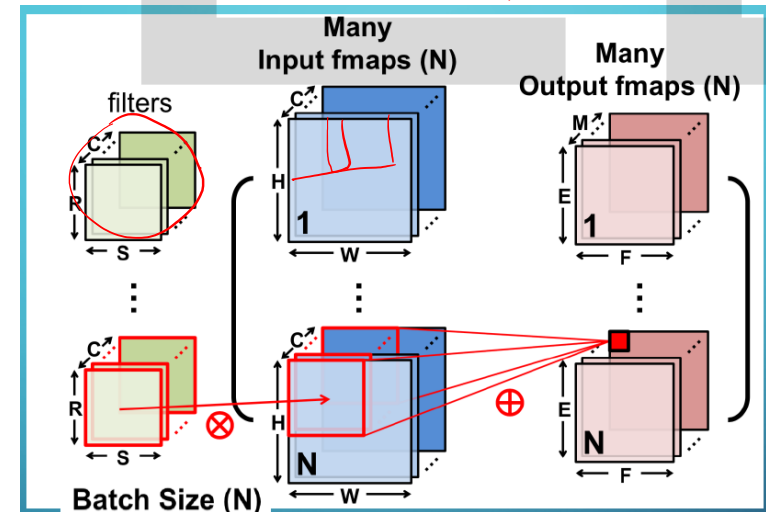
```
for (n=0; n<N; n++) {  
  for (m=0; m<M; m++) {  
    for (x=0; x<F; x++) {  
      for (y=0; y<E; y++) {  
        o[n][m][x][y] = B[m];  
        for (i=0; i<R; i++) {  
          for (j=0; j<S; j++) {  
            for (k=0; k<C; k++) {  
              o[n][m][x][y] += I[n][k][Ux+i][Uy+j] x W[m][k][i][j];  
            }  
          }  
        }  
      }  
    }  
  }  
}
```

convolve a window and apply activation

for each output fmap value

Handwritten annotations:

- N : #images/batch
- M : #filters
- F : dimension of o/p fmap
- E : dimension of filters
- $B[m]$: #channels in i/p fmap
- R : dim. of filters
- S : #channels in i/p fmap
- C : stride
- $Ux+i$: Bias

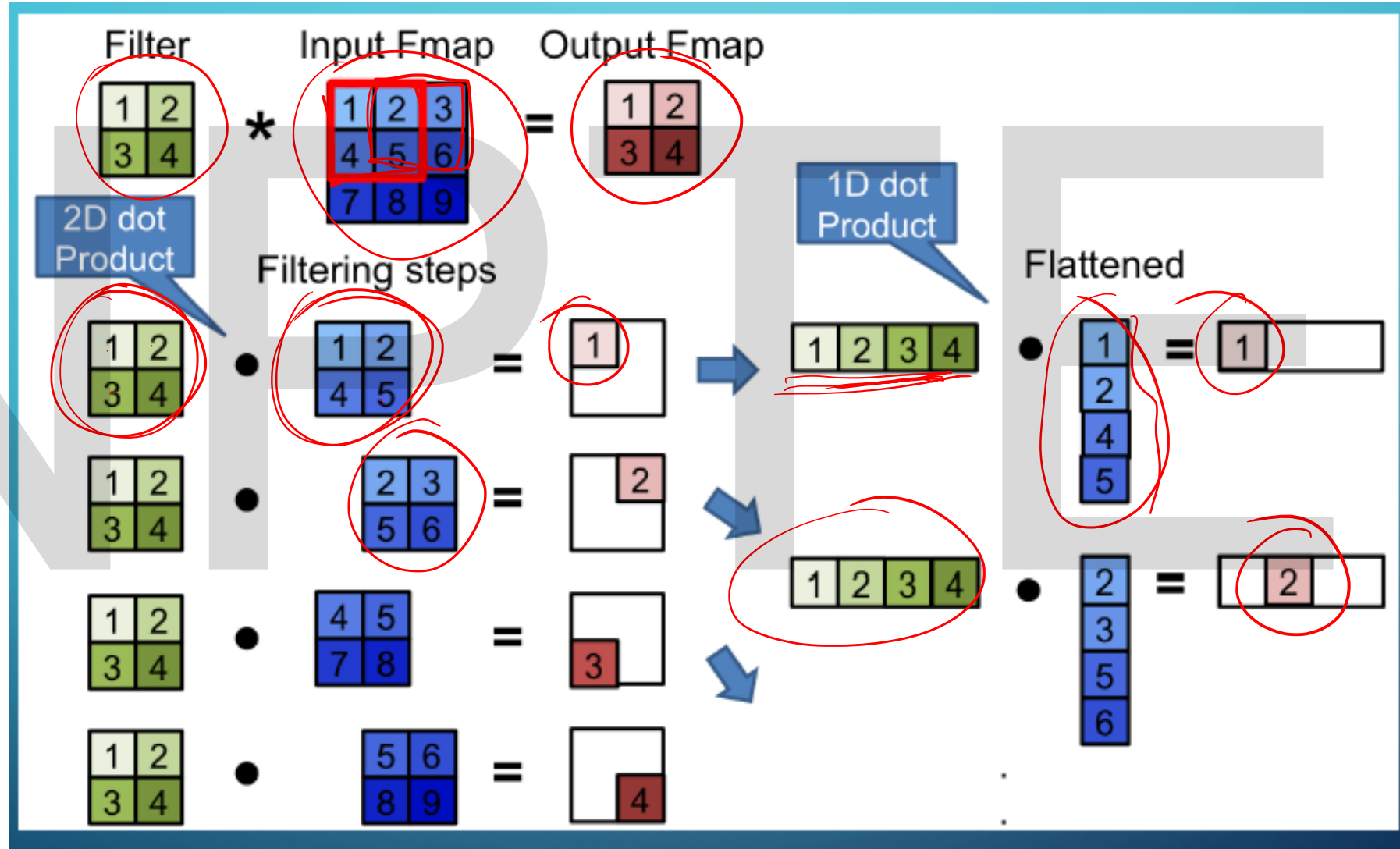


Agenda

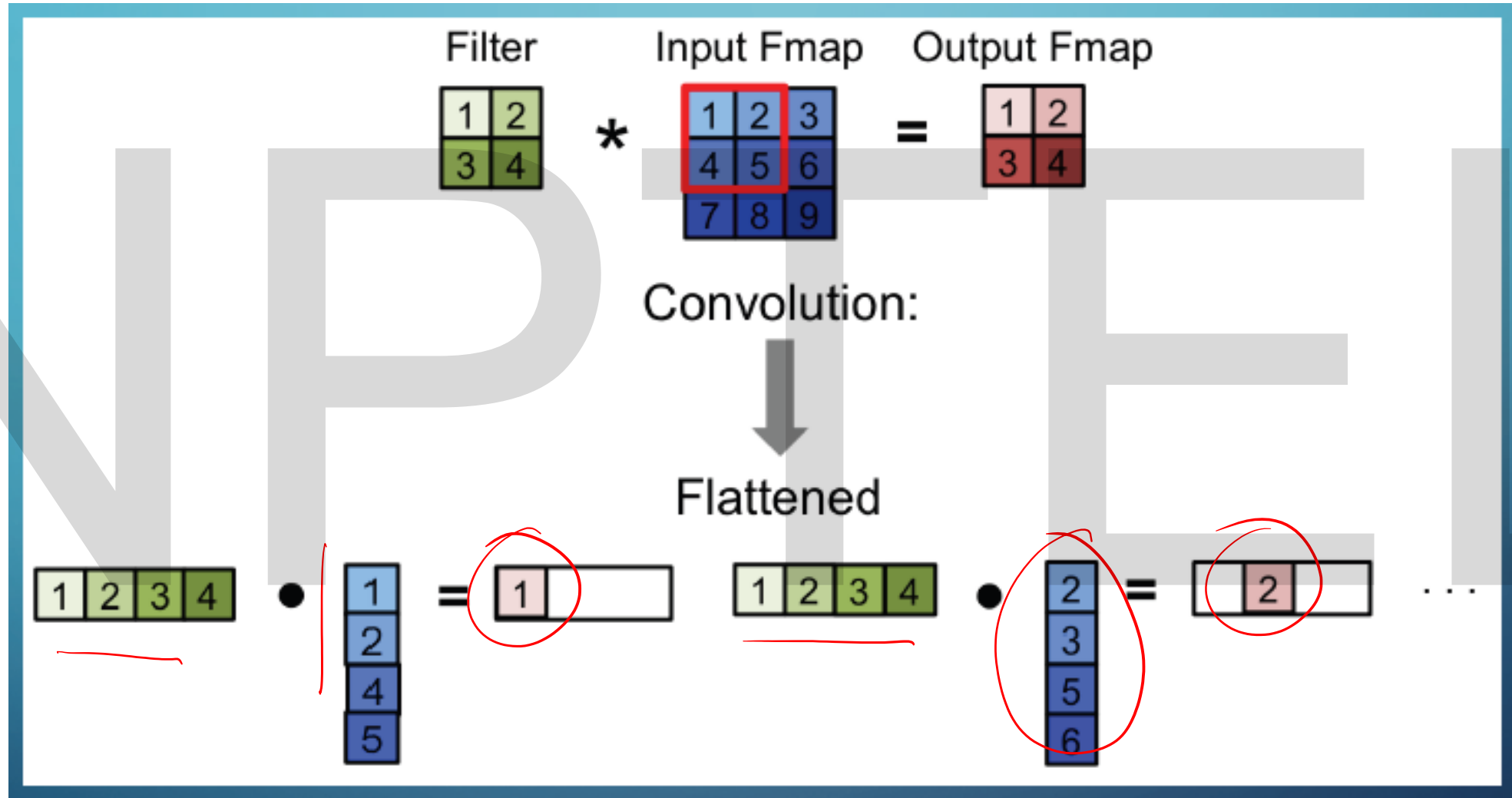
- Recap
- **Convolution computation**

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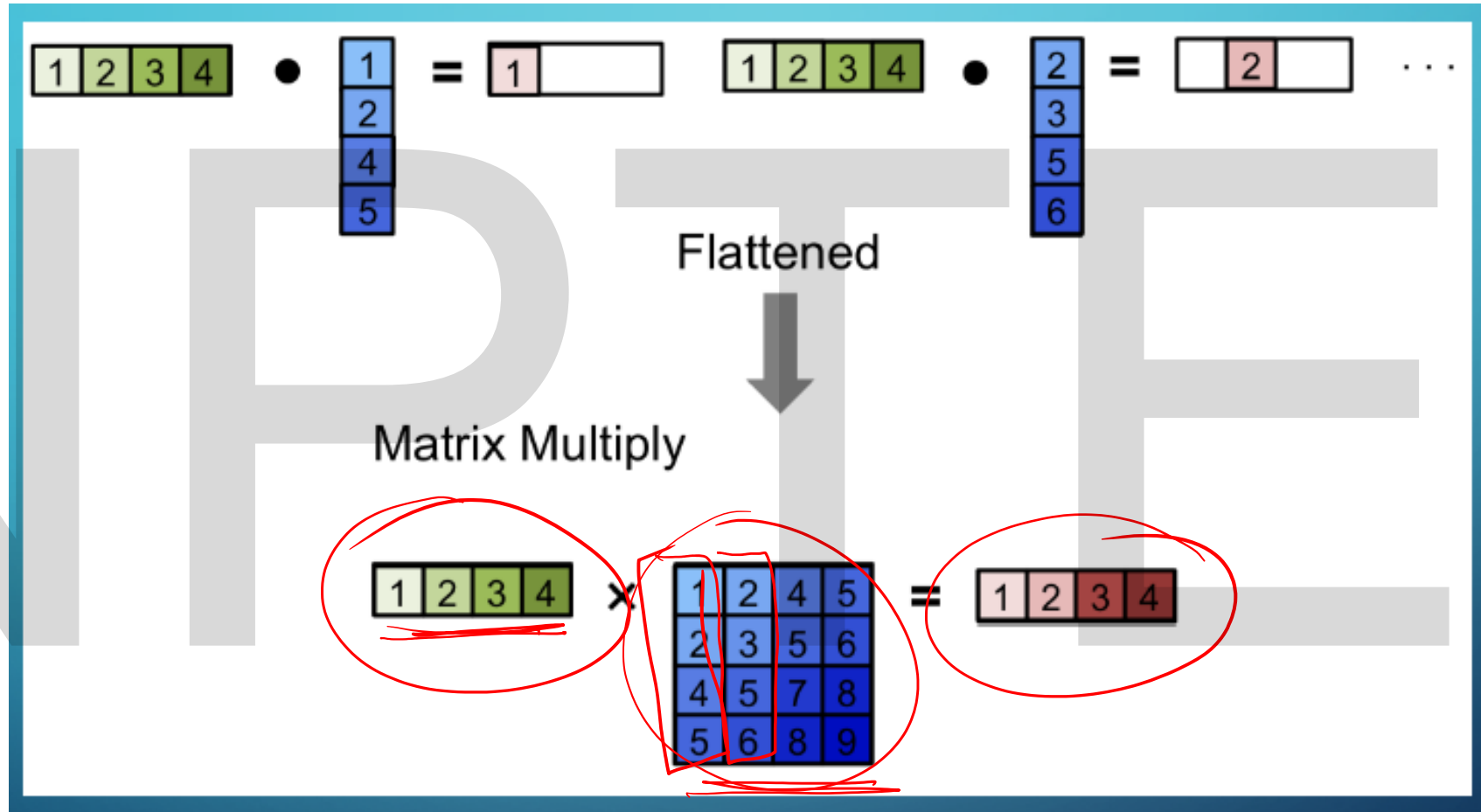
Conv Layer Computation



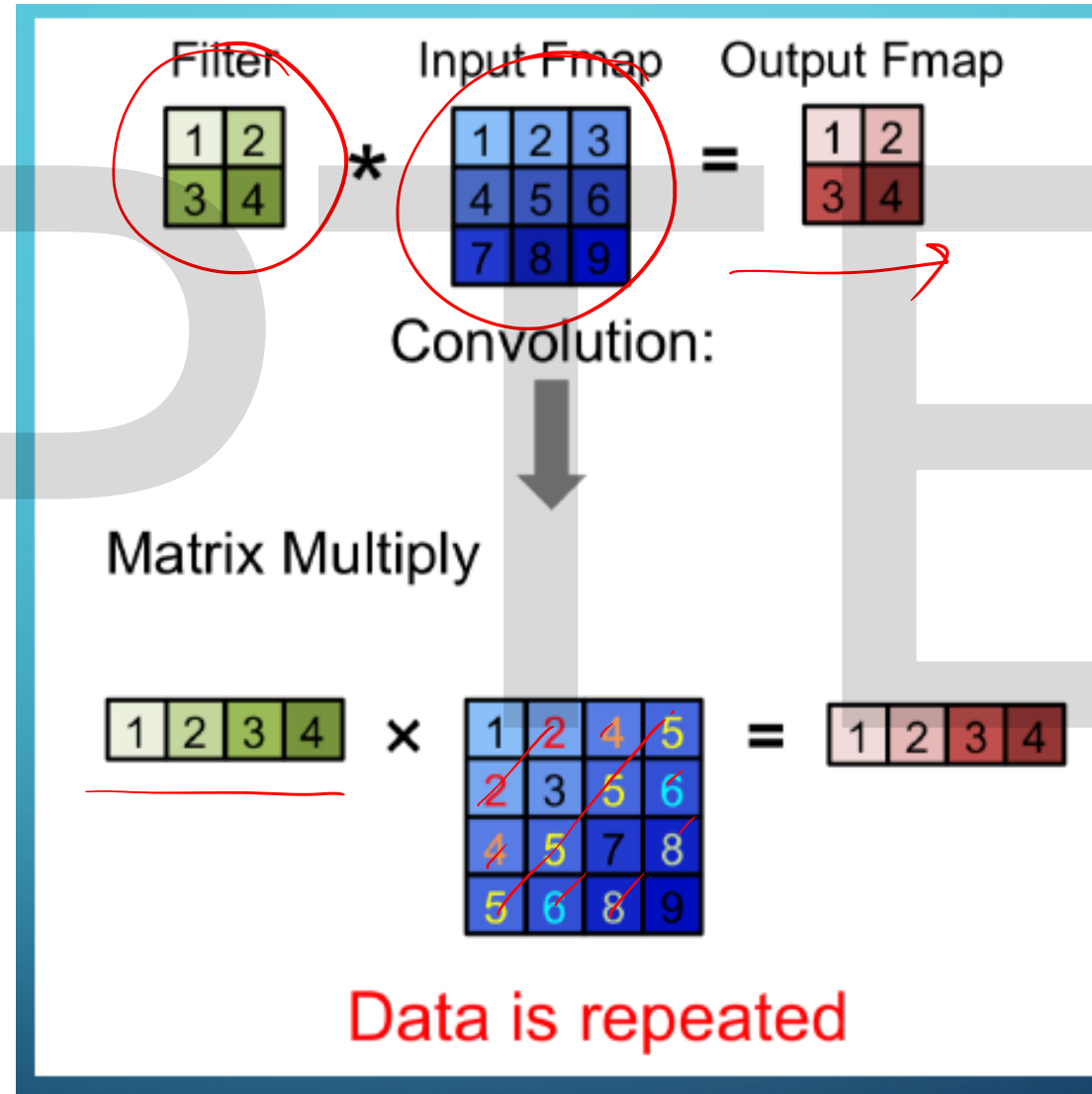
Conv Layer Computation



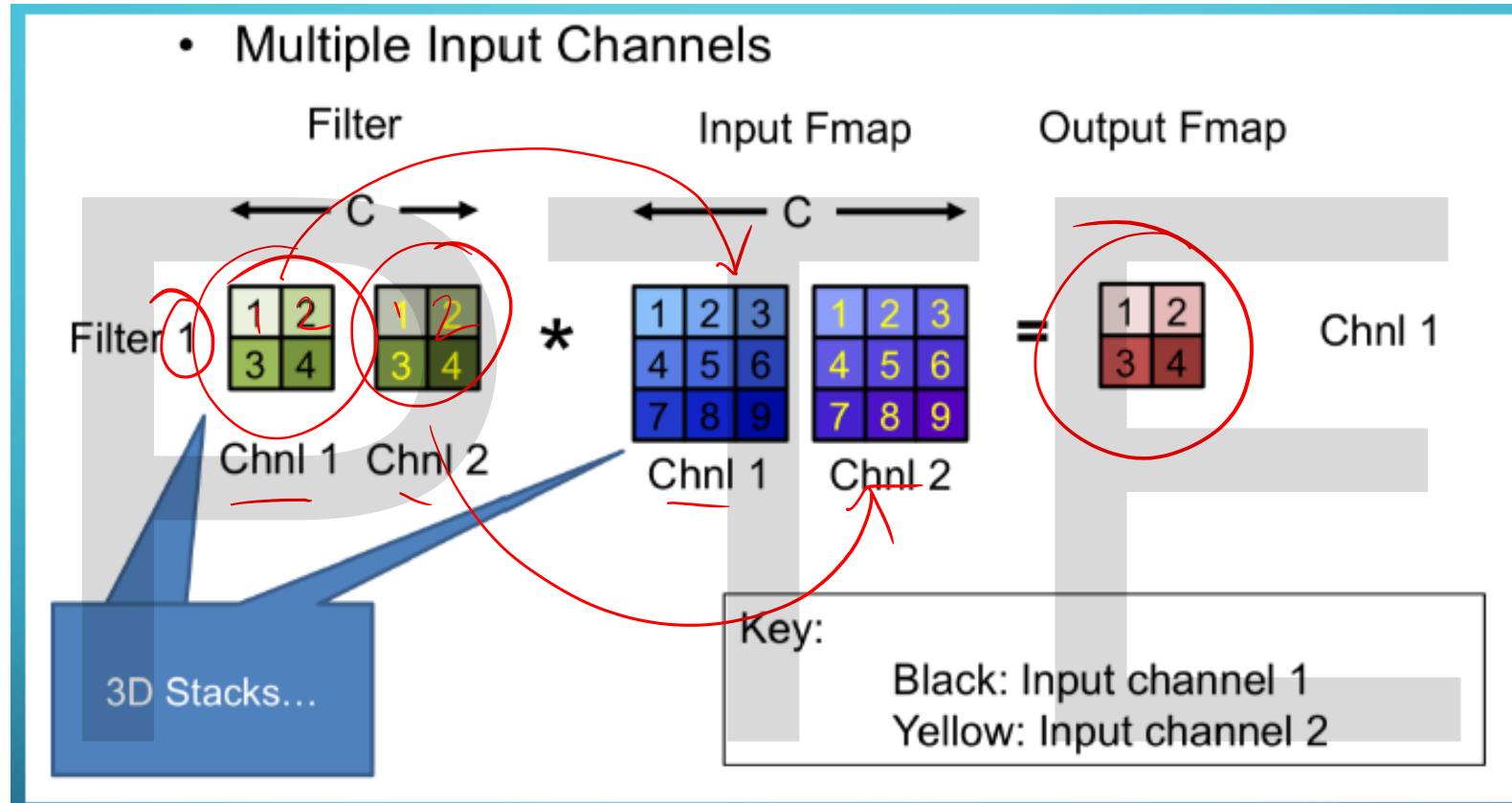
Conv Layer Computation



Conv Layer Computation

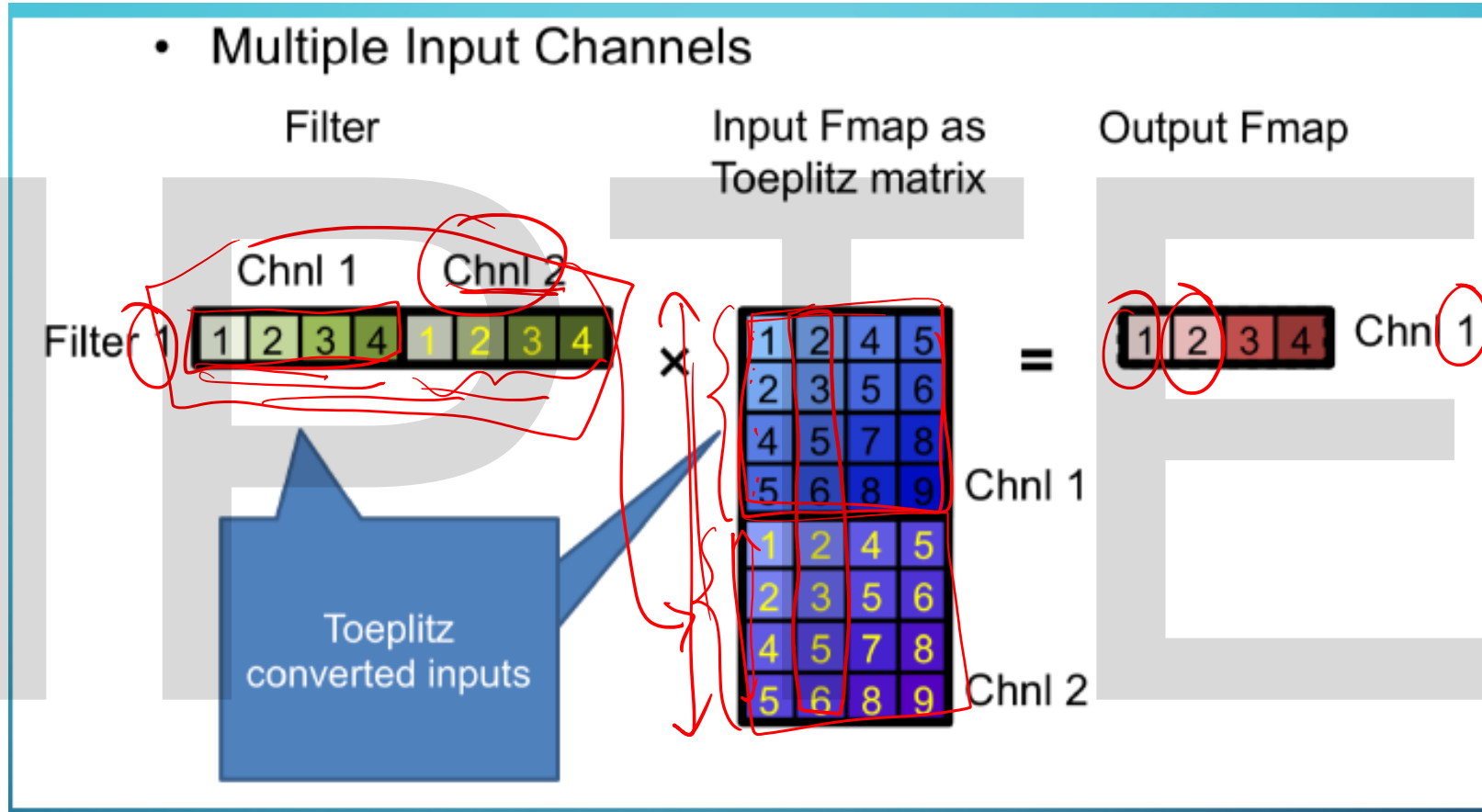


Conv Layer Computation



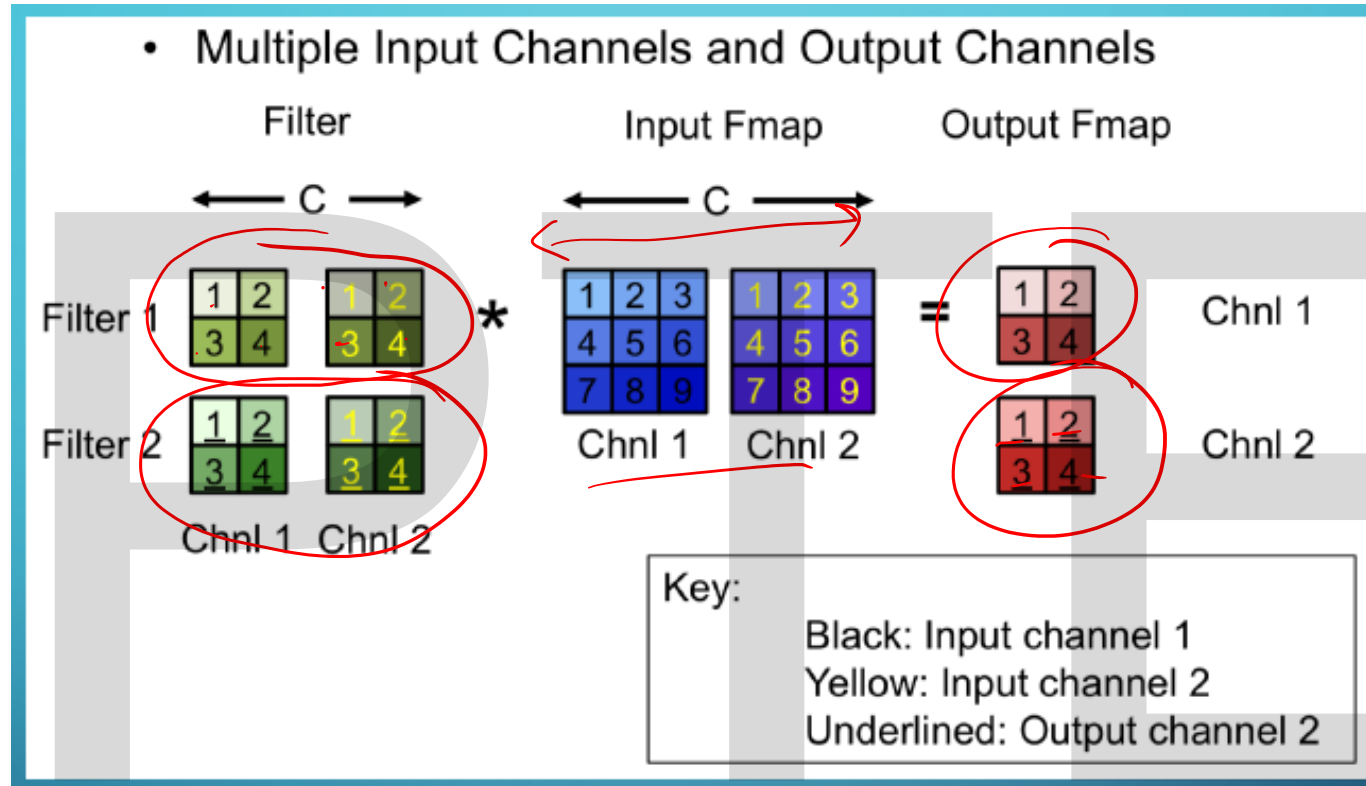
- Both channel 1 and channel 2 in the left are used to generate channel 1 of the output

Conv Layer Computation



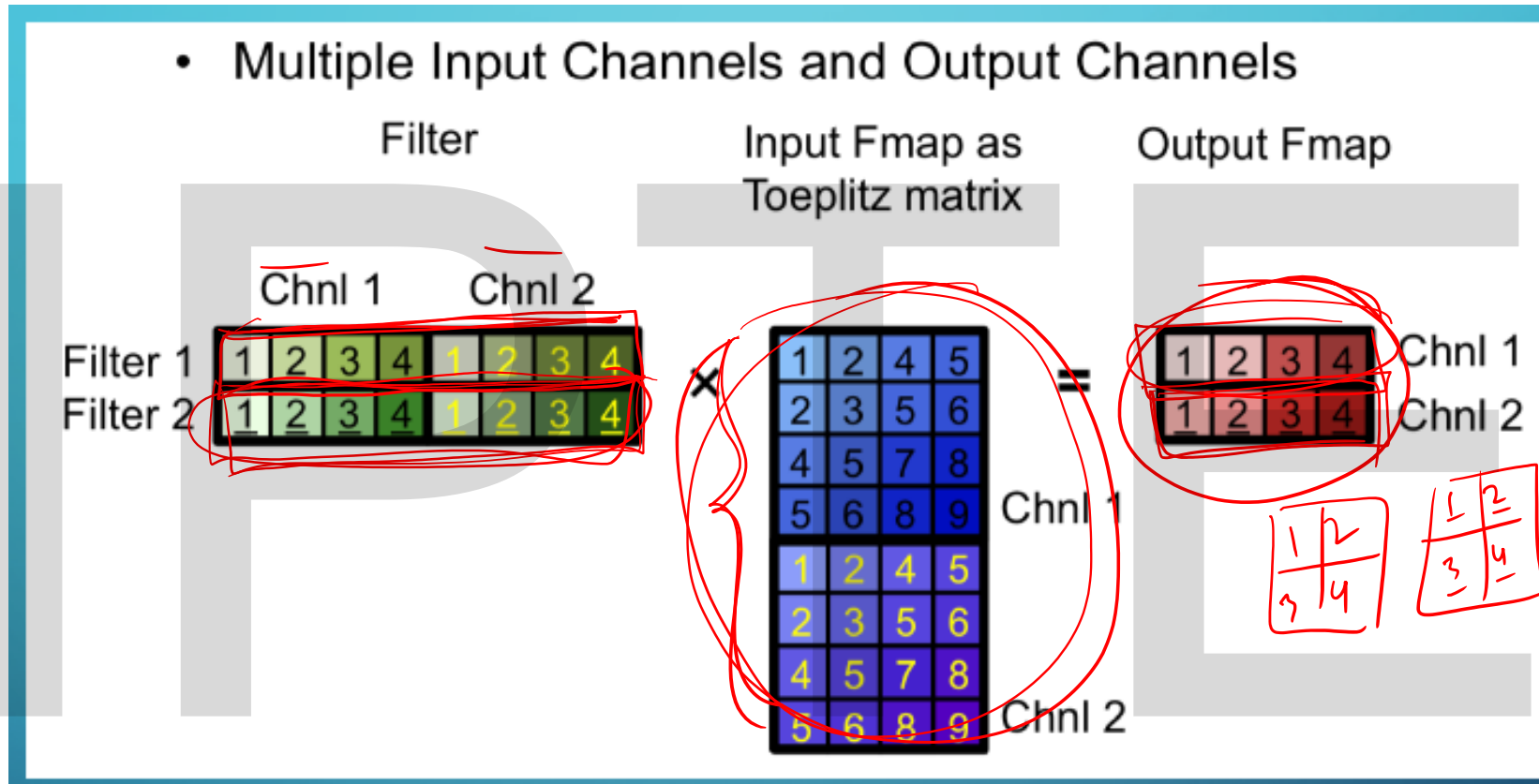
- We can flatten both the filters and the input feature map
 - The computation is vector-matrix multiplication

Conv Layer Computation



- Filter 2 corresponds to (channel 1, channel 2) of input feature map, but channel 2 of output feature map
- Similarly, Filter 1: (channel 1, channel 2) of input and channel 1 of output

Conv Layer Computation

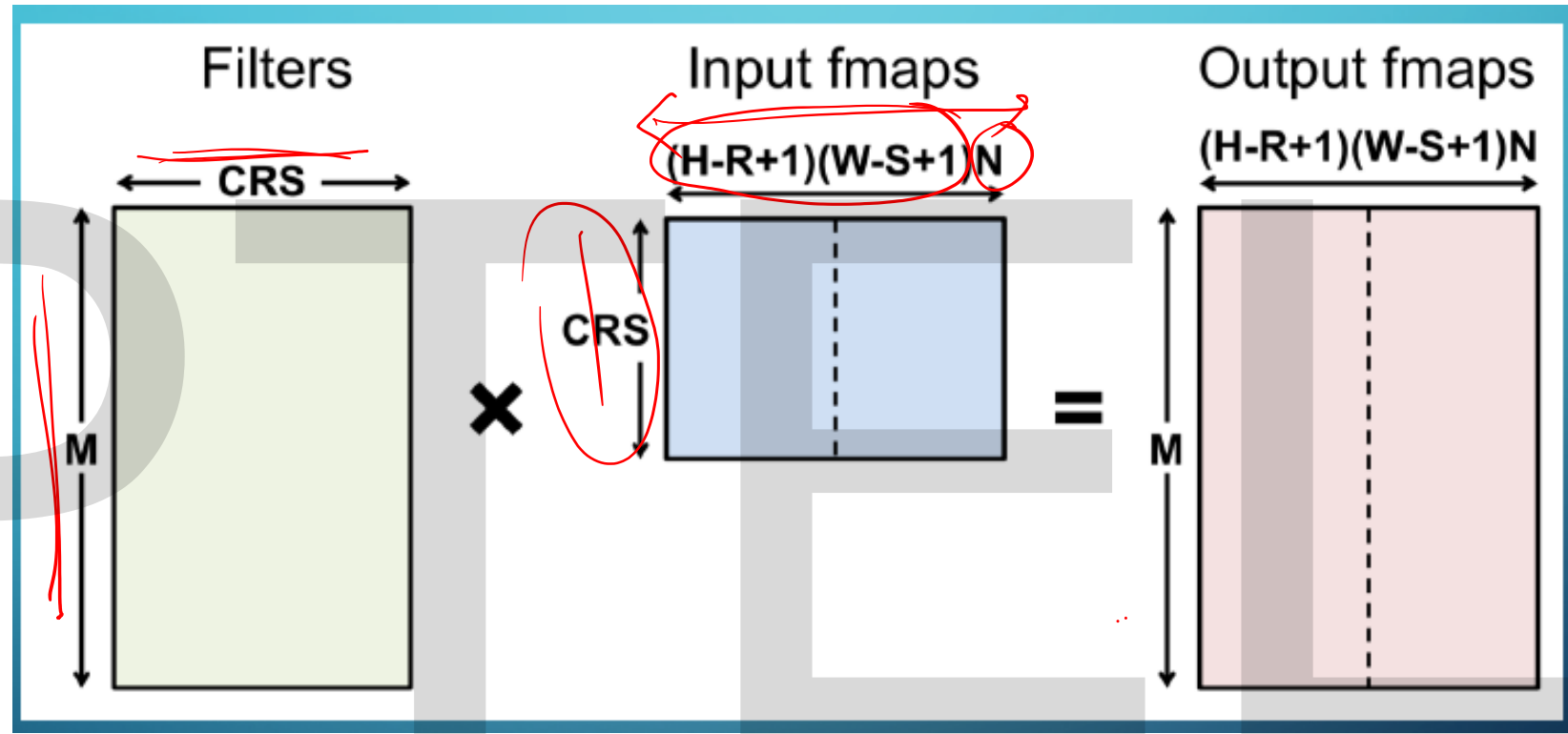


- We can flatten both filters and input feature maps in the same way as before
- Naturally generate flattened output feature maps of two channels

Conv Layer Computation

- What are R, S, C, M?

- R: Height of the filter
- S: Width of the filter
- C: Number of channels
- M: Number of filters





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

